

AI STUDYBUDDY API

AI-Powered Educational Backend

Project Report

Technology Stack

Node.js • Express.js • MongoDB • Gemini 2.5 Flash • JWT • Multer

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A backend system for intelligent study-material processing, personalized learning, AI-generated content, and secure user management.

1. ABSTRACT

AI StudyBuddy API is an AI-powered educational backend designed to support students with personalized and interactive learning services. The system provides secure authentication, study-material upload, AI summarization, flashcard generation, quiz generation, and personalized study-plan creation.

The application uses Node.js and Express.js to implement RESTful APIs, MongoDB to store users and study materials, JWT-based authentication for secure access, and Gemini 2.5 Flash for AI-powered educational features. Uploaded materials can be processed and converted into useful learning resources.

The backend follows role-based access control with student and admin roles. Students can manage their own materials and use AI features, while administrators can view users, delete users and related materials, and monitor system statistics.

The main objective is to reduce manual preparation time and provide a reusable learning assistant through a simple web/API interface.

Keywords: Artificial Intelligence, Study Assistant, Node.js, Express.js, MongoDB, Gemini 2.5 Flash, JWT, REST API.

2. INTRODUCTION

Modern students often need to convert large amounts of study content into concise and useful learning resources. Reading long notes repeatedly can be time-consuming. AI StudyBuddy addresses this problem by providing backend APIs that transform uploaded study materials into summaries, flashcards, quizzes, and study plans.

The system accepts study materials in .txt, .md, and .pdf formats. After authentication, a student can upload material and request an AI operation for that material. The Gemini 2.5 Flash model is used to generate educational outputs based on the uploaded content.

The backend separates responsibilities into controllers, middleware, models, routes, and utility modules. This structure improves maintainability and makes the project easier to extend.

Objectives

- Provide secure registration and login using JWT cookies.
- Allow students to upload and manage study materials.
- Generate AI summaries, flashcards, and MCQ quizzes.
- Create personalized study plans based on user goals.
- Provide admin-only user and statistics management.

3. SYSTEM REQUIREMENTS & TECHNOLOGIES

Technology	Purpose
Node.js	JavaScript runtime for backend execution.
Express.js	Web framework for REST API routes and middleware.
MongoDB	NoSQL database for users and study materials.
Gemini 2.5 Flash	AI engine for summaries, flashcards, quizzes and study plans.
JWT	Authentication and authorization token mechanism.
HTTP-only Cookies	Secure browser-side storage of access and refresh tokens.
Multer	File upload handling and temporary storage.
dotenv	Loads environment variables from the .env file.
PDF/TXT/MD processing	Supports common study-material formats.

Minimum Software Requirements

- Node.js and npm installed.
- MongoDB running locally or a configured MongoDB server.
- A Gemini API key.
- VS Code or another JavaScript development environment.
- Modern web browser or API testing tool such as Postman.

4. SYSTEM ARCHITECTURE

The system follows a layered REST API architecture. Client requests first reach Express routes. Authentication middleware verifies the user session and role where required. Controllers contain application logic, models communicate with MongoDB, and utility modules handle database connection, Gemini requests, and token management.

Layer	Components	Responsibility
Client	Browser / API Client	Sends HTTP requests and displays responses.
Routes	auth, materials, admin	Maps endpoints to controllers.
Middleware	auth, upload	Authentication, authorization and file handling.
Controllers	Auth, Material, Admin	Implements application operations.
Models	User, Material	Defines MongoDB data structures.
Utilities	db, gemini, tokens	Database, AI and JWT helper functions.
Database	MongoDB	Stores persistent application data.

Request Flow: Client → Express Route → Middleware → Controller → Utility/Model → MongoDB or Gemini → JSON Response

5. AUTHENTICATION & ROLE-BASED ACCESS

AI StudyBuddy uses JWT authentication with HTTP-only cookies. Two tokens are used: an access token with a short lifetime and a refresh token with a longer lifetime. The access token is intended for normal authenticated requests, while the refresh token is used to obtain a new access token.

Access token expiry: 15 minutes. Refresh token expiry: 7 days.

Authentication endpoints

Method	Endpoint	Description
POST	/api/auth/register	Create a new user account.
POST	/api/auth/login	Authenticate a user.
POST	/api/auth/refresh	Refresh authentication tokens.
POST	/api/auth/logout	Clear authentication cookies.

Role-Based Access Control

- **Student:** Can access personal study materials and AI learning features.
- **Admin:** Can access administrator routes for user and system management.

The auth middleware protects private routes, while the admin-only middleware checks whether the authenticated account has the required admin role.

6. STUDY MATERIAL MANAGEMENT

Students can upload study materials using multipart form-data. Supported formats are .txt, .md, and .pdf. Multer handles the uploaded file and temporary storage. Material information is then associated with the authenticated user in MongoDB.

Method	Endpoint	Function
POST	/api/materials/upload	Upload a study material with optional title.
GET	/api/materials/	List the logged-in user's materials.
GET	/api/materials/:id	View one material.
DELETE	/api/materials/:id	Delete one material.

Typical Upload Process

- User logs in and receives secure authentication cookies.
- User selects a supported study file.
- Client sends multipart/form-data to the upload endpoint.
- Multer processes the file.
- The material record is stored in MongoDB.
- The API returns a success response and material information.

The uploads folder is intended as temporary file storage, while MongoDB stores the application metadata and ownership information.

7. AI-POWERED LEARNING FEATURES

Gemini 2.5 Flash provides the main AI functionality. The material controller sends suitable study content and instructions to the Gemini helper. The generated result is returned through the API as a structured response.

Endpoint	Input	Output / Purpose
POST /:id/summarize	Material ID	Concise AI-generated summary.
POST /:id/flashcards	count	Flashcards for revision.
POST /:id/quiz	count	Multiple-choice questions.
POST /:id/study-plan	goal, hoursPerDay, days	Personalized study plan.

Example flashcard request body: { count: 5 }

Example quiz request body: { count: 5 }

Example study-plan request body: { goal, hoursPerDay, days }

These services allow one uploaded source to become multiple learning resources, supporting revision, self-testing, and structured preparation.

8. ADMIN MODULE, DATABASE & SECURITY

Admin routes provide management functions that are restricted to users with the admin role.

Method	Endpoint	Description
GET	/api/admin/users	List all users.
DELETE	/api/admin/users/:id	Delete a user and their materials.
GET	/api/admin/stats	Return total user and material counts.

Database

MongoDB is used as the persistent database. The project contains User and Material models. A local database can use the connection string `mongodb://localhost:27017/ai-studybuddy`.

Security Measures

- JWT-based authentication.
- HTTP-only cookies for token storage.
- Separate access and refresh tokens.
- Role-based authorization for admin endpoints.
- Environment variables for secrets and API keys.
- User ownership checks for material operations.

The actual production deployment should additionally use HTTPS, strong secrets, secure cookie settings, input validation, rate limiting, and appropriate file-size/type restrictions.

9. SETUP, API SUMMARY & CONCLUSION

Setup Steps

1. Install dependencies using **npm install**.
2. Create a **.env** file and add PORT, MONGO_URI, JWT_ACCESS_SECRET, JWT_REFRESH_SECRET, GEMINI_API_KEY, and NODE_ENV.
3. Start MongoDB.
4. Run the backend using **node index.js**.
5. Test the endpoints using a browser-based client, frontend application, or API testing tool.

Environment Variables

Variable	Example / Purpose
PORT	5000
MONGO_URI	mongodb://localhost:27017/ai-studybuddy
JWT_ACCESS_SECRET	Strong private secret
JWT_REFRESH_SECRET	Different strong private secret
GEMINI_API_KEY	Gemini API credential
NODE_ENV	development

Conclusion

AI StudyBuddy API combines secure backend development with AI-assisted education. It provides a modular foundation for authentication, study-material management, AI content generation, and administration. The architecture can later be extended with a frontend dashboard, progress tracking, notifications, analytics, and additional learning features.

Team

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END OF PROJECT REPORT